

✓ New Section

```
import pandas as pd
```

```
df = pd.read_csv('data.csv')
df.info()
df.describe()
df.fillna(method='ffill', inplace=True)
df.drop_duplicates(inplace=True)
df['price'] = df['price'].astype(float)
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 54329 entries, 0 to 54328
Data columns (total 11 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   ad_id       54329 non-null   object
1   type        54329 non-null   object
2   price       54329 non-null   float64
3   area        54329 non-null   float64
4   bedrooms    54329 non-null   object
5   bathrooms   54329 non-null   object
6   level       54329 non-null   object
7   furnished   42448 non-null   object
8   rent        54329 non-null   object
9   city        54329 non-null   object
10  region      54329 non-null   object
dtypes: float64(2), object(9)
memory usage: 4.6+ MB
<ipython-input-6-8f16f4e20661>:6: FutureWarning: DataFrame.fillna with 'method' is deprecated and will raise in a future version. Use of
df.fillna(method='ffill', inplace=True)
```

```
from ydata_profiling import ProfileReport
```

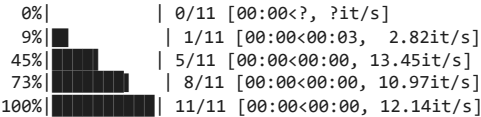
```
profile = ProfileReport(df, title="Data Cleaning Report", explorative=True)
profile.to_notebook_iframe()
profile.to_file("data_cleaning_report.html")
from google.colab import files
```

```
files.download("data_cleaning_report.html")
```



Summarize dataset: 100%

24/24 [00:02<00:00, 9.80it/s, Completed]



Generate report structure: 100%

1/1 [00:04<00:00, 4.27s/it]

Render HTML: 100%

1/1 [00:00<00:00, 1.85it/s]

Data Cleaning Report

Overview Variables Interactions Correlations Missing values Sample

A simple visualization of nullity by column.

Sample

First rows

Last rows

	ad_id	type	price	area	bedrooms	bathrooms	level	furnished	rent	city	
0	Ad id 194281336	apartment	3790000.0	207.0	3	3	2	NaN	no	giza	6th of
1	Ad id 194198684	apartment	870000.0	155.0	3	3	4	no	no	cairo	new h
2	Ad id 191102186	apartment	1546400.0	170.0	3	2	9	no	no	cairo	zahraa
3	Ad id 193951860	apartment	1300000.0	270.0	4	3	ground	no	no	sharqia	10th c
4	Ad id 194064010	duplex	5888000.0	387.0	4	4	ground	no	no	cairo	new c
5	Ad id 194021448	apartment	950000.0	104.0	2	1	7	no	no	cairo	nasr c
6	Ad id 194294715	apartment	2100000.0	160.0	3	2	1	no	no	cairo	mosta
7	Ad id 194320699	apartment	2550000.0	215.0	3	2	ground	no	no	giza	6th of
8	Ad id 194316553	apartment	3994232.0	160.0	3	3	ground	no	no	cairo	new c
9	Ad id 194217278	duplex	2310000.0	280.0	3	3	ground	no	no	giza	6th of

Report generated by YData.

Export report to file: 100%

1/1 [00:00<00:00, 47.98it/s]

!pip install pandera



Collecting pandera

Downloading pandera-0.23.1-py3-none-any.whl.metadata (18 kB)
Requirement already satisfied: numpy>=1.24.4 in /usr/local/lib/python3.11/dist-packages (from pandera) (2.0.2)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.11/dist-packages (from pandera) (24.2)
Requirement already satisfied: pandas>=2.1.1 in /usr/local/lib/python3.11/dist-packages (from pandera) (2.2.2)
Requirement already satisfied: pydantic in /usr/local/lib/python3.11/dist-packages (from pandera) (2.11.3)
Requirement already satisfied: typeguard in /usr/local/lib/python3.11/dist-packages (from pandera) (4.4.2)
Collecting typing_inspect>=0.6.0 (from pandera)
Downloading typing_inspect-0.9.0-py3-none-any.whl.metadata (1.5 kB)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.11/dist-packages (from pandas>=2.1.1->pandera) (2.8.2)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.11/dist-packages (from pandas>=2.1.1->pandera) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.11/dist-packages (from pandas>=2.1.1->pandera) (2025.2)
Collecting mypy_extensions>=0.3.0 (from typing_inspect>=0.6.0->pandera)
Downloading mypy_extensions-1.0.0-py3-none-any.whl.metadata (1.1 kB)
Requirement already satisfied: typing_extensions>=3.7.4 in /usr/local/lib/python3.11/dist-packages (from typing_inspect>=0.6.0->pandera)
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.11/dist-packages (from pydantic->pandera) (0.7.0)
Requirement already satisfied: pydantic-core==2.33.1 in /usr/local/lib/python3.11/dist-packages (from pydantic->pandera) (2.33.1)
Requirement already satisfied: typing-inspection>=0.4.0 in /usr/local/lib/python3.11/dist-packages (from pydantic->pandera) (0.4.0)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.11/dist-packages (from python-dateutil>=2.8.2->pandas>=2.1.1->pandera)

```
Downloading pandera-0.23.1-py3-none-any.whl (264 kB)  
264.2/264.2 kB 17.1 MB/s eta 0:00:00  
Downloading typing_inspect-0.9.0-py3-none-any.whl (8.8 kB)  
Downloading mypy_extensions-1.0.0-py3-none-any.whl (4.7 kB)  
Installing collected packages: mypy-extensions, typing_inspect, pandera  
Successfully installed mypy-extensions-1.0.0 pandera-0.23.1 typing_inspect-0.9.0
```

```
import pandera as pa  
from pandera import Column, DataFrameSchema  
  
schema = DataFrameSchema({  
    "ad_id": Column(str),  
    "type": Column(str),  
    "price": Column(float, nullable=False),  
    "area": Column(float, nullable=False),  
    "bedrooms": Column(str, nullable=True),  
    "bathrooms": Column(str, nullable=True),  
    "level": Column(str, nullable=True),  
    "furnished": Column(str, nullable=True),  
    "rent": Column(str, nullable=True),  
    "city": Column(str),  
    "region": Column(str)  
})  
  
validated_df = schema.validate(df)  
print("✅ All data is valid according to schema.")
```